

University Interscholastic League
2022 – 2023 Elementary Number Sense Test C

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.
Stop – Wait for Signal!

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| <p>(1) $5 \times 9 =$ _____</p> <p>(2) $277 + 722 =$ _____</p> <p>(3) $79 - 43 =$ _____</p> <p>(4) $1206 \div 6 =$ _____</p> <p>(5) $264 \div 11 =$ _____</p> <p>(6) $17 + 19 + 18 =$ _____</p> <p>(7) $4 \times 7 \times 5 =$ _____</p> <p>(8) $876 - 676 =$ _____</p> <p>(9) $51 \times 8 =$ _____</p> <p>*(10) $2022 + 4981 =$ _____</p> <p>(11) $83 \times 11 =$ _____</p> <p>(12) Which digit is in the ten-thousandths place in 34976.15820? _____</p> <p>(13) $18 \times 12 =$ _____</p> <p>(14) 78630.4728 rounded to the ten's place is _____</p> <p>(15) What is the remainder for $82525 \div 3$? _____</p> <p>(16) The number of even whole numbers between 6 and 17 is _____</p> <p>(17) $8 \times 10^2 + 3 \times 10^1 + 2 \times 10^{-2} =$ _____ (decimal)</p> <p>(18) $17 \times 7 - 17 \times 3 =$ _____</p> <p>(19) MMXXIII = _____ (Arabic Numeral)</p> | <p>*(20) $335 \times 179 =$ _____</p> <p>(21) $18 - 8 \div 2 =$ _____</p> <p>(22) $9 + 11 + 13 + 15 + 17 =$ _____</p> <p>(23) $\frac{4}{5} =$ _____ percent</p> <p>(24) $1\frac{1}{4}$ days = _____ hours</p> <p>(25) $\frac{11}{18} + \frac{5}{18} =$ _____</p> <p>(26) $29 \times 31 =$ _____</p> <p>(27) 0.15 = _____ common fraction</p> <p>(28) If 12 ♥ costs 48¢ then 8 ♥ cost _____ ¢</p> <p>(29) $\frac{7}{16} \times 48 =$ _____</p> <p>*(30) $777 \times 449 =$ _____</p> <p>(31) 900 nickels = _____ dimes</p> <p>(32) The product of the two biggest primes smaller than 10 is _____</p> <p>(33) 6 quarts = _____ ounces</p> <p>(34) $2250 \div 25 =$ _____</p> <p>(35) $55\frac{5}{9}\% =$ _____ common fraction</p> <p>(36) The LCM of 12 and 16 is _____</p> <p>(37) $99 \times 97 =$ _____</p> |
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- (38) The ratio in inches of 3 feet to 2 yards is _____
- (39) $4\frac{1}{4} \times 8\frac{1}{4} =$ _____ (mixed number)
- *(40) $376 \times 479 =$ _____
- (41) The side for a cube with volume 64 cm^3 is _____ cm
- (42) $21^2 =$ _____
- (43) The perimeter of a square with area 289 m^2 is _____ m
- (44) $2^4 + 2^3 =$ _____
- (45) $10\frac{3}{5} \div \frac{1}{5} =$ _____
- (46) $\sqrt{625} =$ _____
- (47) What is the perimeter of a right triangle with legs 6 and 8? _____
- (48) $44 \times 16 =$ _____
- (49) If $x = 18$, then $18 + 3x =$ _____
- *(50) $23 + 230 + 2023 + 2022 =$ _____
- (51) What is the number, **k**, in the sequence:
64, 32, 16, 8, **k**, 2, ...? _____
- (52) $11\frac{5}{8} - 5\frac{7}{8} =$ _____ (mixed number)
- (53) If the area of a circle is 625π , what is the diameter of the circle? _____
- (54) What is the volume of a rectangular box that measures 8" by 15" by 5"? _____ in^3
- (55) $213 \text{ (base 5)} =$ _____ (base 10)
- (56) What whole number cubed plus twenty-five equals fifty-two? _____
- (57) A triangle has sides of 14, 17, and a semiperimeter of 25. What is the third side? _____
- (58) If set $A = \{2, 4, 6, \dots, 26\}$ and set $B = \{3, 6, 12, 22\}$, then the number of elements in $A \cap B$ is _____
- (59) What is the perimeter of the equilateral triangle with a side length of $12\frac{1}{3}$? _____
- *(60) 188 days = _____ hours
- (61) $67 \text{ (base 8)} =$ _____ (base 2)
- (62) $-4^3 \times (2) =$ _____
- (63) Two fair dice are thrown. What is the probability that the sum of the two sides showing is 7? _____
- (64) 0.5 square miles = _____ acres
- (65) $53^2 =$ _____
- (66) $4^6 \div 7$ has remainder of _____
- (67) How many vertices does a cube have? _____
- (68) If $3x - 12 < 36$, then $x <$ _____
- (69) $\frac{4}{9} + \frac{9}{4} = 2 +$ _____
- *(70) $125 \times \sqrt{255} =$ _____
- (71) The additive inverse of $1\frac{3}{7}$ is _____
- (72) The area of a square with diagonal 22 is _____
- (73) If $6\frac{1}{2}\%$ of x is $19\frac{1}{2}\%$ of 12, then $x =$ _____
- (74) $(32) \div (-8) \times (4) =$ _____
- (75) $9^2 + 18^2 =$ _____
- (76) $\left(12\frac{1}{2}\right)^2 - \left(7\frac{1}{2}\right)^2 =$ _____
- (77) What is the area of a trapezoid with bases 15, 18 and height 10? _____
- (78) $286 \times 14 =$ _____
- (79) $59^2 + 59 =$ _____
- *(80) $49 \times 50 \times 51 =$ _____

2022 – 2023 University Interscholastic League Elementary Number Sense Test C – Key

(1) 45	*(20) 56967 – 62963	(38) $\frac{1}{2}$; .5	(59) 37
(2) 999	(21) 14		*(60) 4287 – 4737
(3) 36	(22) 65	(39) $35\frac{1}{16}$	(61) 110111
(4) 201	(23) 80		(62) -128
(5) 24	(24) 30	*(40) 171099 – 189109	(63) $\frac{1}{6}$
(6) 54	(25) $\frac{8}{9}$	(41) 4	(64) 320
(7) 140	(26) 899	(42) 441	(65) 2809
(8) 200		(43) 68	(66) 1
(9) 408	(27) $\frac{3}{20}$	(44) 24	(67) 8
*(10) 6653 – 7353	(28) 32	(45) 53	(68) 16
(11) 913	(29) 21	(46) 25	
(12) 2	*(30) 331430 – 366316	(47) 24	(69) $\frac{25}{36}$
(13) 216	(31) 450	(48) 704	*(70) 1897 – 2095
(14) 78630	(32) 35	*(50) 4084 – 4512	(71) $-1\frac{3}{7}$; $-\frac{10}{7}$
(15) 1	(33) 192	(51) 4	(72) 242
(16) 5	(34) 90	(52) $5\frac{3}{4}$	(73) 36
(17) 830.02	(35) $\frac{5}{9}$	(53) 50	(74) -16
(18) 68	(36) 48	(54) 600	(75) 405
(19) 2023	(37) 9603	(55) 58	(76) 100
		(56) 3	(77) 165
		(57) 19	(78) 4004
		(58) 3	(79) 3540
			*(80) 118703 – 131197

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.